

Clinic Management System

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1- Explanation: Rationale for Design Choices & Implementation Notes

Design Rationale

The Clinic Management System database was designed to efficiently manage patient records, doctor information, clinic departments, and appointment scheduling.

The database was normalized to reduce data redundancy and improve data consistency. Each major entity in the system was represented as a separate table:

- **Department** stores information about medical departments.
- **Clinic** stores clinic information and is linked to a department.
- **Doctor** stores doctor information and is associated with a department.
- **Patient** stores personal information about patients.
- **Appointment** stores appointment details and connects patients with doctors.

Primary keys were used to uniquely identify each record, while foreign keys were implemented to maintain referential integrity between related tables.

The Appointment table serves as the central table because it connects patients and doctors while storing appointment details, costs, diagnoses, and status information.

Implementation Notes

- MySQL was used as the database management system.
- Foreign key constraints were implemented to prevent invalid references between tables.
- Data types were selected according to the nature of the stored information:
 - INT for IDs
 - VARCHAR for names and addresses
 - DATE for birth dates and appointment dates
 - TIME for appointment times
 - DECIMAL for appointment costs
- A view was created to simplify access to cardiology clinic information.
- Sample data was inserted into all tables to test functionality and query execution.

2-Conceptual Design

1. Department Entity

- DepartmentID (Primary Key)
- DepartmentName

2. Clinic Entity

- ClinicID (Primary Key)
- ClinicName
- Address
- Department (Foreign Key referencing DepartmentID)

3. Doctor Entity

- DoctorID (Primary Key)
- DoctorName
- PhoneNumber
- Address
- Department (Foreign Key referencing DepartmentID)

4. Patient Entity

- PatientID (Primary Key)
- PatientName
- PhoneNumber
- Address
- BirthDate
- Job

5. Appointment Entity

- AppointmentID (Primary Key)
- AppointmentDate
- StartTime
- EndTime

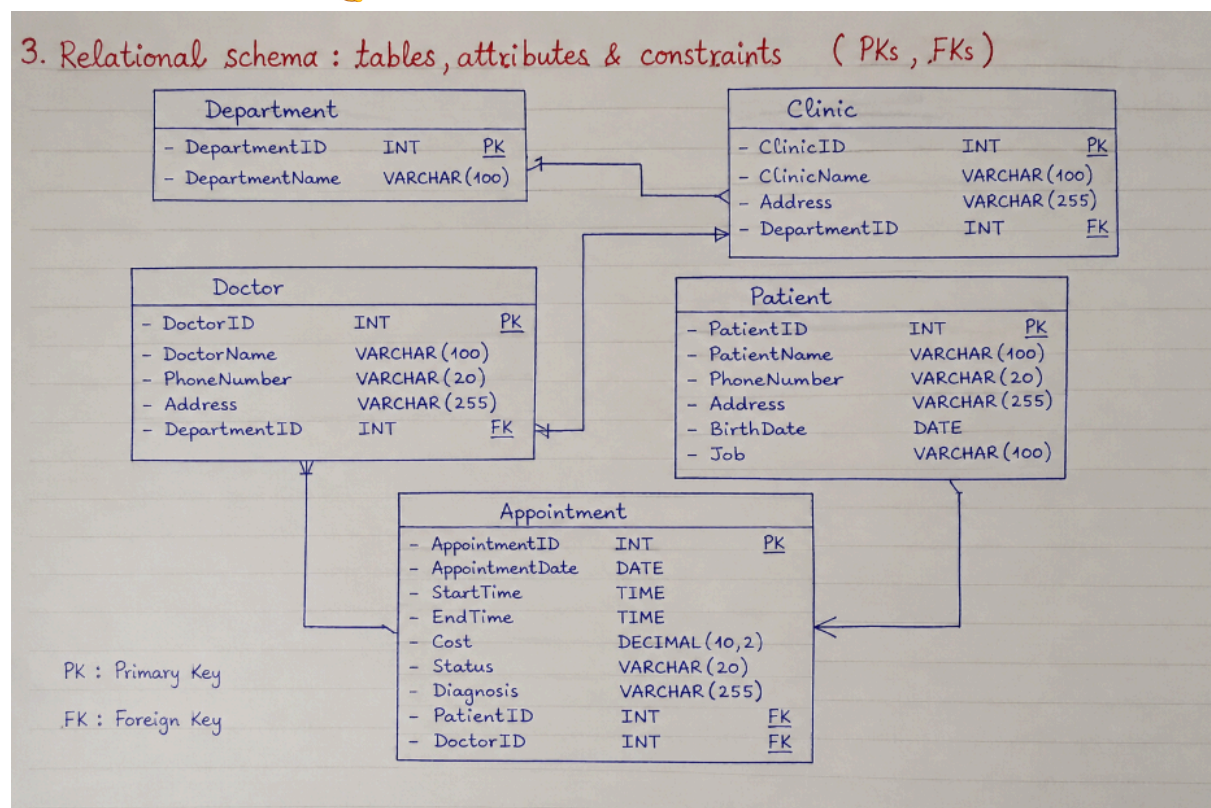
- Cost
- Status
- Diagnosis
- PatientID (Foreign Key referencing PatientID)
- DoctorID (Foreign Key referencing DoctorID)

Relationships Between Entities

- Department → Clinic : One department can have many clinics (1:M).
- Department → Doctor : One department can have many doctors (1:M).
- Patient → Appointment : One patient can have many appointments (1:M).
- Doctor → Appointment : One doctor can have many appointments (1:M)

3-Relational Scheme 👍

3. Relational schema : tables, attributes & constraints (PKs, FKs)



4- DDL & DML Statements to create the tables - Create view and insert



```
1 • CREATE DATABASE ClinicManagement;
2 • USE ClinicManagement;
3 • CREATE TABLE Department (
4     DepartmentID INT PRIMARY KEY,
5     DepartmentName VARCHAR(100) NOT NULL
6 );
```

```
15 • CREATE TABLE Patient (
16     PatientID INT PRIMARY KEY,
17     PatientName VARCHAR(100) NOT NULL,
18     PhoneNumber VARCHAR(20),
19     Address VARCHAR(255),
20     BirthDate DATE,
21     Job VARCHAR(100)
22 );
23 • CREATE TABLE Doctor (
24     DoctorID INT PRIMARY KEY,
25     DoctorName VARCHAR(100) NOT NULL,
26     PhoneNumber VARCHAR(20),
27     Address VARCHAR(255),
28     DepartmentID INT,
29     FOREIGN KEY (DepartmentID)
30         REFERENCES Department(DepartmentID)
31 );
```

```
32 • CREATE TABLE Appointment (
33     AppointmentID INT PRIMARY KEY,
34     AppointmentDate DATE,
35     StartTime TIME,
36     EndTime TIME,
37     Cost DECIMAL(10,2),
38     Status ENUM('Scheduled', 'In Progress', 'Postponed'),
39     Diagnosis VARCHAR(255),
40     PatientID INT,
41     DoctorID INT,
42     FOREIGN KEY (PatientID)
43         REFERENCES Patient(PatientID),
44     FOREIGN KEY (DoctorID)
45         REFERENCES Doctor(DoctorID)
46 );
```

```
3 • CREATE TABLE Department (
4     DepartmentID INT PRIMARY KEY,
5     DepartmentName VARCHAR(100) NOT NULL
6 );
7 • CREATE TABLE Clinic (
8     ClinicID INT PRIMARY KEY,
9     ClinicName VARCHAR(100) NOT NULL,
10    Address VARCHAR(255),
11    DepartmentID INT,
12    FOREIGN KEY (DepartmentID)
13        REFERENCES Department(DepartmentID)
14 );
```

Output					
Action Output					
#	Time	Action	Message	Duration / Fetch	
✓ 14	15:52:35	CREATE VIEW CardiologyClinics AS SELECT ClinicName, Address FROM Clinic WHERE DepartmentID = 1	0 row(s) affected	0.000 sec	
✓ 15	15:52:35	SELECT * FROM Department LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec	
✓ 16	15:52:35	SELECT * FROM Clinic LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec	
✓ 17	15:52:35	SELECT * FROM Doctor LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec	
✓ 18	15:52:35	SELECT * FROM Patient LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec	

```
24 • INSERT INTO Doctor VALUES
25     (201, 'Ahmed Hassan', '01011111111', 'Cairo', 1),
26     (202, 'Mona Ali', '01022222222', 'Giza', 2),
27     (203, 'Khaled Samy', '01033333333', 'Alexandria', 3),
28     (204, 'Nour Emad', '01044444444', 'Cairo', 4),
29     (205, 'Omar Adel', '01055555555', 'Giza', 5),
30     (206, 'Sara Mohamed', '01066666666', 'Cairo', 6),
31     (207, 'Youssef Many', '01077777777', 'Tanta', 7),
```

```
13 • INSERT INTO Clinic VALUES
14     (101, 'Heart Care Clinic', 'Nasr City', 1),
15     (102, 'Advanced Cardio Center', 'Maadi', 1),
16     (103, 'Brain Health Clinic', 'Heliopolis', 2),
17     (104, 'Skin Beauty Clinic', '6th October', 3),
18     (105, 'Kids Medical Center', 'Dokki', 4),
19     (106, 'Bone & Joint Clinic', 'Mohandessin', 5),
20     (107, 'Vision Eye Center', 'Zamalek', 6),
21     (108, 'ENT Specialist Clinic', 'Shoubra', 7),
22     (109, 'Urology Care Center', 'Faisal', 8),
```

```
46 • INSERT INTO Appointment VALUES
47     (301, '2025-01-15', '10:00:00', '10:30:00', 500, 'Scheduled', 'Fatty liver', 12527, 201),
48
49     (302, '2025-02-20', '11:00:00', '11:30:00', 700, 'In Progress', 'Hypertension', 12528, 201),
50
51     (303, '2024-11-05', '09:00:00', '09:30:00', 450, 'Postponed', 'Fatty liver', 12529, 202),
52
53     (304, '2023-06-18', '13:00:00', '13:30:00', 300, 'Scheduled', 'Migraine', 12527, 202),
54
55     (305, '2025-03-22', '14:00:00', '14:30:00', 600, 'Scheduled', 'Skin Allergy', 12530, 203),
```

```
66 • CREATE VIEW CardiologyClinics AS
67     SELECT ClinicName, Address
68     FROM Clinic
69     WHERE DepartmentID = 1;
70 • SELECT * FROM Department;
71 • SELECT * FROM Clinic;
72 • SELECT * FROM Doctor;
73 • SELECT * FROM Patient;
```

SET LIMIT FOR
Workbench

Action Output					
#	Time	Action	Message	Duration / Fetch	
✓ 3	15:49:28	CREATE TABLE Department (DepartmentID INT PRIMARY KEY, DepartmentName VARCHAR(100) NO...	0 row(s) affected	0.031 sec	
✓ 4	15:49:28	CREATE TABLE Clinic (ClinicID INT PRIMARY KEY, ClinicName VARCHAR(100) NOT NULL, Addres...	0 row(s) affected	0.047 sec	
✓ 5	15:49:28	CREATE TABLE Patient (PatientID INT PRIMARY KEY, PatientName VARCHAR(100) NOT NULL, P...	0 row(s) affected	0.032 sec	
✓ 6	15:49:28	CREATE TABLE Doctor (DoctorID INT PRIMARY KEY, DoctorName VARCHAR(100) NOT NULL, Ph...	0 row(s) affected	0.062 sec	
✓ 7	15:49:28	CREATE TABLE Appointment (AppointmentID INT PRIMARY KEY, AppointmentDate DATE, StartTim...	0 row(s) affected	0.063 sec	

Tables and data results:

DepartmentID	DepartmentName
1	Cardiology
2	Neurology
3	Dermatology
4	Pediatrics
5	Orthodontics

PatientID	PatientName	PhoneNumber	Address	BirthDate	Job
12527	Menna Mohamed	01111111111	Cairo	2003-05-10	Student
12528	Ali Mahmoud	01122222222	Giza	1998-09-15	Engineer
12529	Sara Ahmed	01133333333	Alexandria	2000-01-20	Teacher
12530	Youssef Adel	01144444444	Cairo	1995-07-12	Designer
12531	Maha Samir	01155555555	Giza	1999-11-08	Doctor
12532	Mohamed Nasser	01166666666	Mansoura	1997-03-14	Accountant
12533	Nadine Fathy	01177777777	Tanta	2001-06-21	Lawyer
12534	Karim Ashraf	01188888888	Cairo	1996-12-01	Programmer
12535	Aya Mostafa	01199999999	Alexandria	2002-09-17	Student
12536	Hany Ibrahim	01112312312	Giza	1994-04-25	Architect
NULL	NULL	NULL	NULL	NULL	NULL

AppointmentID	AppointmentDate	StartTime	EndTime	Cost	Status	Diagnosis	PatientID	DoctorID
301	2025-01-15	10:00:00	10:30:00	500.00	Scheduled	Fatty liver	12527	201
302	2025-02-20	11:00:00	11:30:00	700.00	In Progress	Hypertension	12528	201
303	2024-11-05	09:00:00	09:30:00	450.00	Postponed	Fatty liver	12529	202
304	2023-06-18	13:00:00	13:30:00	300.00	Scheduled	Migraine	12527	202
305	2025-03-22	14:00:00	14:30:00	600.00	Scheduled	Skin Allergy	12530	203
306	2025-04-10	10:00:00	10:20:00	400.00	Scheduled	Eye Infection	12531	206
307	2025-04-15	11:00:00	11:25:00	550.00	In Progress	Ear Infection	12532	207
308	2025-05-01	12:00:00	12:30:00	650.00	Scheduled	Kidney Stones	12533	208

ClinicID	ClinicName	Address	DepartmentID
101	Heart Care Clinic	Nasr City	1
102	Advanced Cardio Center	Maadi	1
103	Brain Health Clinic	Heliopolis	2
104	Skin Beauty Clinic	6th October	3
105	Kids Medical Center	Dokki	4
106	Bone & Joint Clinic	Mohandessin	5
107	Vision Eye Center	Zamalek	6
108	ENT Specialist Clinic	Shoubra	7
109	Urology Care Center	Faisal	8
110	Digestive Health Clinic	New Cairo	10
NULL	NULL	NULL	NULL

DoctorID	DoctorName	PhoneNumber	Address	DepartmentID
201	Ahmed Hassan	01011111111	Cairo	1
202	Mona Ali	01022222222	Giza	2
203	Khaled Samy	01033333333	Alexandria	3
204	Nour Emad	01044444444	Cairo	4
205	Omar Adel	01055555555	Giza	5
206	Sara Mohamed	01066666666	Cairo	6
207	Youssef Hany	01077777777	Tanta	7
208	Mariam Essam	01088888888	Mansoura	8
209	Hossam Nabil	01099999999	Alexandria	9
210	Rania Tarek	01012345678	Cairo	10
NULL	NULL	NULL	NULL	NULL

Now the trigger:

```
1 DELIMITER $$
2
3 • CREATE TRIGGER CheckAppointmentCost
4 BEFORE INSERT ON Appointment
5 FOR EACH ROW
6 BEGIN
7     IF NEW.Cost < 0 THEN
8         SIGNAL SQLSTATE '45000'
9         SET MESSAGE_TEXT = 'Appointment cost cannot be negative';
10    END IF;
11 END$$
12
13 DELIMITER ;
```

19 15:52:35 SELECT * FROM Appointment LIMIT 0, 1000 10 row(s) returned

20 16:30:26 EXPLAIN INSERT INTO Clinic VALUES (101,'Heart Care Clinic','Nasr City',1), (102,'Advanced Cardio Center',... OK

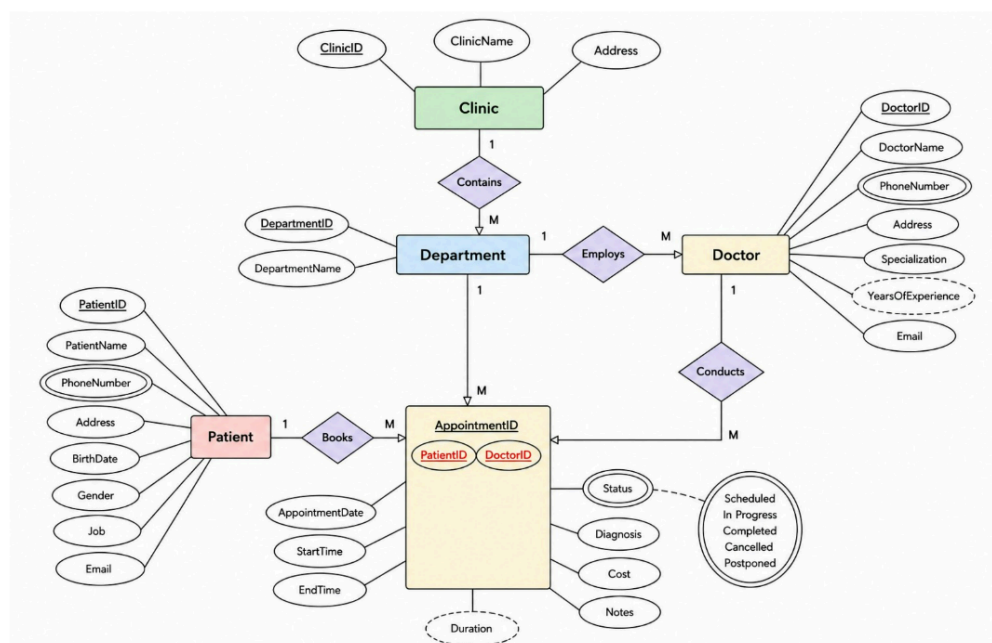
21 16:30:26 EXPLAIN FORMAT=JSON INSERT INTO Clinic VALUES (101,'Heart Care Clinic','Nasr City',1), (102,'Advance... OK

22 18:14:24 CREATE TRIGGER CheckAppointmentCost BEFORE INSERT ON Appointment FOR EACH ROW BEGIN ... 0 row(s) affected

Purpose

- Ensures data integrity.
- Prevents invalid appointment costs from being inserted.
- Reduces data entry errors.

5-The ER Diagram Model



Thank you..